

Unit 2: Exploring Two Variable Data
HW 1 – Two Categorical Variables

Name: _____

1) The table shows results of a poll, asking adults whether they were looking forward to the Super Bowl game, looking forward to the commercials, or didn't plan to watch.

	Male	Female	Total
Game	279	200	479
Commercials	81	156	237
Won't Watch	132	160	292
Total	492	516	1008

a) Display the data with a side-by-side bar chart.

b) Display the data with a segmented bar graph.

c) Based on your investigation, would you agree or disagree that there is an association between a person's sex and their interest in the Super Bowl?

2) A public opinion survey explored the relationship between age and support for increasing the minimum wage. The results are summarized in the two-way table below.

	<i>Yes</i>	<i>No</i>	<i>No Opinion</i>	<i>Total</i>
<i>18 to 30</i>	25	20	5	50
<i>31 to 60</i>	20	35	20	75
<i>Over 60</i>	55	15	5	75
<i>Total</i>	100	70	30	200

a) Give an example of a joint relative frequency and interpret it in context.

b) Give an example of a marginal relative frequency and interpret it in context.

c) Give an example of a conditional relative frequency and interpret it in context.

3) The table below shows the 3 most popular social media platforms and the distribution of the ages on those platforms. The data is based on MAU (monthly active users). A random sample of 100 active users from each of the platforms is taken and their age was recorded.

		Social Media Platform			
		Facebook	YouTube	WhatsApp	Total
Age Group	0 – 17	6	23	17	46
	18 – 26	28	40	18	86
	27 – 35	37	30	28	95
	36 – 49	23	5	17	45
	50 and older	6	2	20	28
		100	100	100	300

Create a mosaic plot of the data above. Show ALL work and calculations used to create the plot.